

Republic Of Yemen
University Of Aden - Faculty Of Engineering
Odd Semester Examination - Academic Year 2022/2023

Class: B2IT& B2ITE

Date: 2/2/2023

Name: _____

Subject: Analog Electronics

Time: 2 hours

Reg. No.: _____

Answer question (1) (compulsory) and any three questions.

Question (1)

ترفق ورقة الأسئلة بملف الإجابة.

Choose the correct answer.

a) A transistor can be operated as an electronic switch in

i. Cutoff

ii. Saturation

iii. Both of them

b) The ratio of dc collector current to dc emitter current in a BJT is

i. α

ii. β

iii. γ

c) Add a dc level to an ac voltage called

i. Diode clampers

ii. Diode clippers

iii. Diode rectifier

d) Rectifier is an electronic circuit that converts

i. DC to AC

ii. AC to DC

iii. AC to AC

e) In half wave rectifier, If the peak voltage is 20 V and load resistance is 2 K Ω ,
than the output DC current is

i. 0.01A

ii. 3.2 mA

iii. 9.3 mA

f) A center-tapped full wave rectifier connected to a transformer whose each
secondary coil has $V_{rms}=14$ V, $r_f=40\Omega$, $R_L=1$ k Ω then the V_{dc} is

i. 12.11V

ii. 12.11mV

iii. 19.79 V

g) In cutoff, V_{CE} is

i. 0 V

ii. Equal to V_{CC}

iii. Equal to V_C

h) The most heavily doped of the three semiconductor regions of a BJT is

- i. Collector ii. Base iii. Emitter

i) An example of the simplest form of diode clipper is

- i. Half-wave rectifier ii. Bridge rectifier iii. Center tap rectifier

j) The unit of efficiency is

- i. Watt ii. Volt iii. None of them

2# Put true or false and correct the mistakes.

- i. No matter what type of configuration a transistor is used, the base-to-emitter voltage is zero volt if the transistor is in the "on" state. ()
- ii. The percentage of DC component in rectifier output is called ripple factors ()
- iii. I_B is always greater than I_C ()
- iv. If I_C is 50 times larger than I_B than β is 100. ()
- v. If β times the value of R_E is at least 5 times the value of R_2 , the approximate approach can be applied with a high degree of accuracy. ()

Question (2)

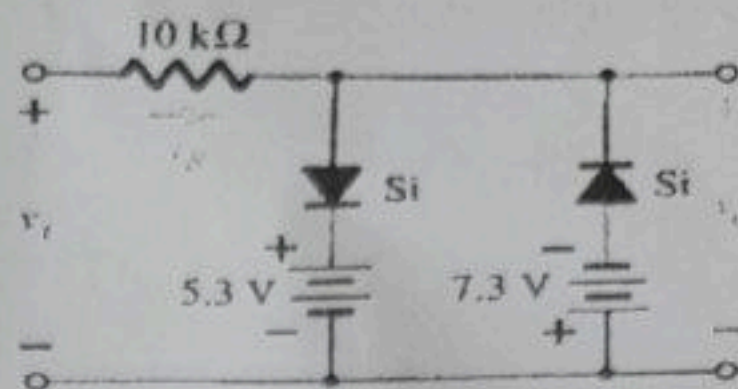
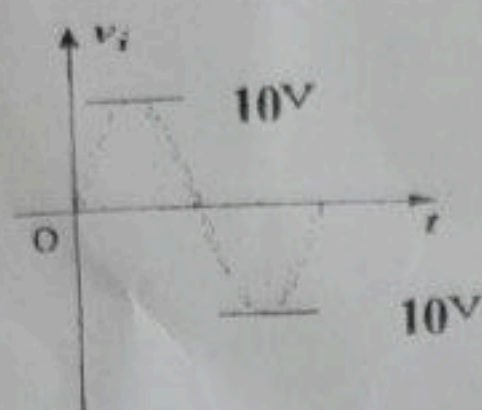
A center-tapped full wave rectifier connected to a transformer whose input r.m.s. voltage is 240 V. Assume the internal resistances of the diode and load resistance are 60 and 2k Ω , respectively and turn ratio 10:2.

a) Draw the center-tapped full wave rectifier circuit.

b) Determine: I_{dc} , V_{dc} , Rectification efficiency .

Question (3)

- Enumerate the DC biasing methods of BJT?
- What is the purpose of emitter resistor added to the DC bias?
- In figure shown below Determine the output voltage waveform.



Question (4)

- From clamping circuit with silicon diode in Fig.1 Determine the output voltage waveform.
- From Fig.2 what is the name of this circuit? Determine β , R_C , R_B , V_{CE} , V_E . (Where $I_C = 3.2 \text{ mA}$).

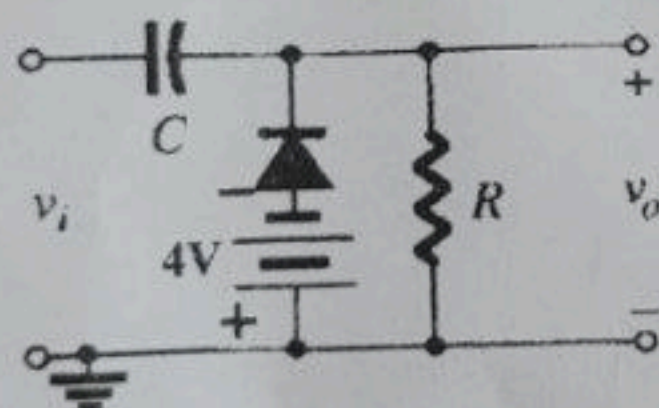
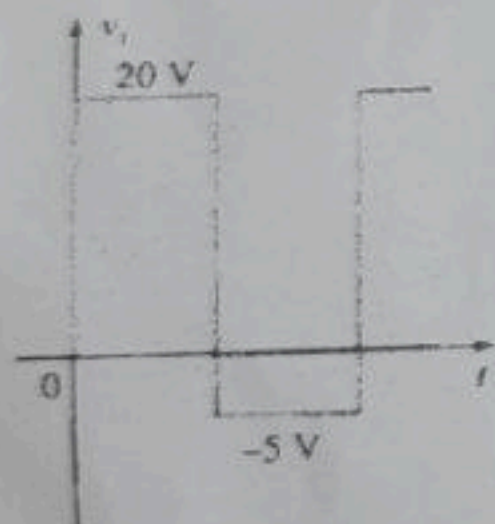


Fig.1

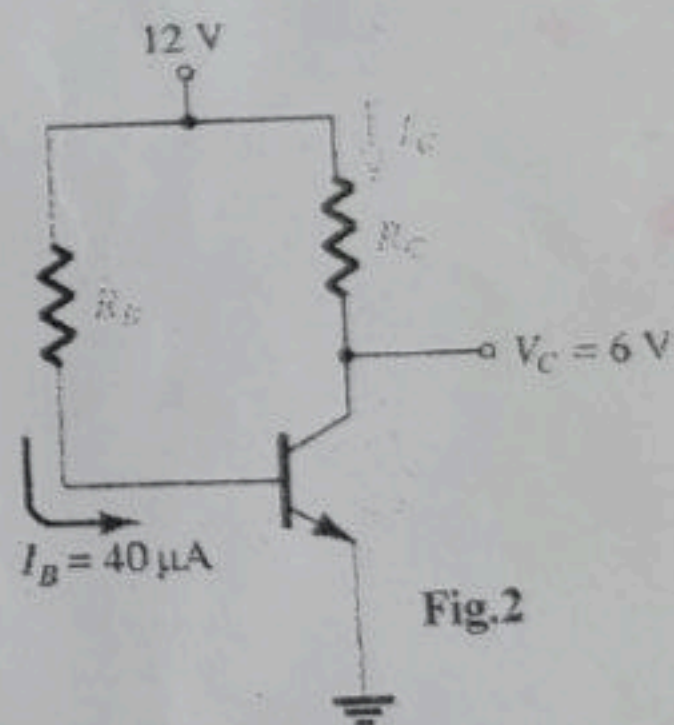


Fig.2

Question (5)

For the voltage-divider bias configuration of Fig.3

- Draw the Thevenin equivalent circuit
- Determine: I_B , I_C , V_{CE} , V_C .

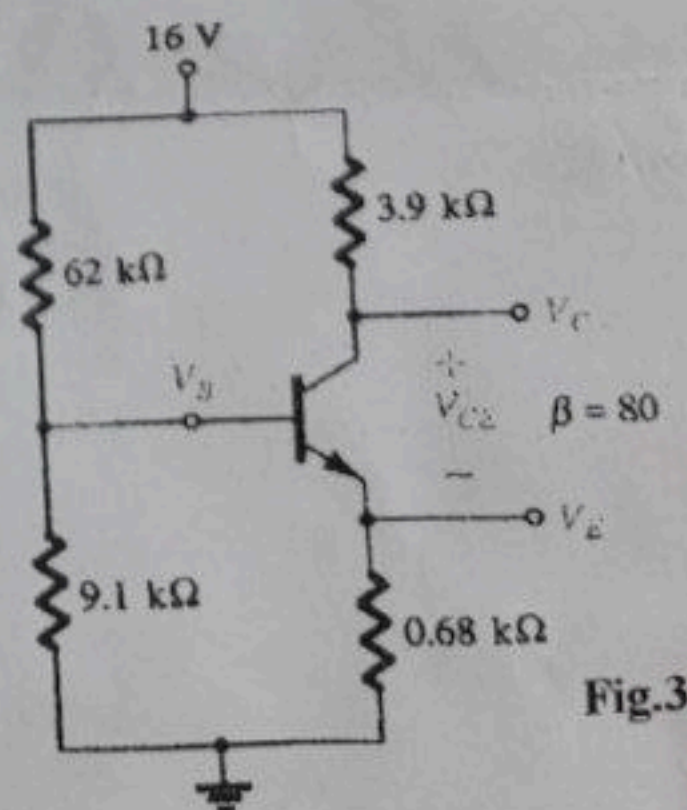


Fig.3